**FIN — Release 10.34**

# Research Programs P394–P410

Formal boundary, contact geometry, noisy operational design, and an executable Jordan-sampling specification

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: 0009-0002-0909-3613

31 July 2026

Publication — Preprint; Version 1.0.0

**Exact contact geometry.** The frozen degree-eleven dual has nine critical points in $(0, 1)$, isolated by exact rational Sturm arithmetic. Every one of the seven P366 primal atoms has strictly positive complementary slack; the displayed primal/dual pair is near-contact, not exact-contact.
**Operational completion.** A frozen JSON protocol, event schema, validator, hashes, and 50,000 synthetic events implement the Jordan Sampling Realization. The software passes; physical evidence is explicitly not admitted.
**Boundary.** No selector, full legacy-to-strict bridge, dimensional source, role transfer, calibrated laboratory realization, Standard Model, gravity, or Theory-of-Everything closure is claimed.

**Repository.** `https://github.com/hyconiek/Fractal-Nadsoliton-Theory`

# Publication statement

This monograph reports every executed Program P394–P410 and recommends Programs P411–P427. Mathematical proof, proof-kernel reflection, numerical evidence, conditional operational construction, and external empirical admission are kept distinct. The legacy/strict split and current strict-core guardrails remain binding.

# Contents

# Abstract


This monograph executes FIN Research Programs P394–P410 without reopening exhausted selector, generic bridge, role-transfer, dimensional-source, or total-action loops.

P394 reflects the complete depth-fourteen Bernstein subdivision algorithm through Lean. Exact de Casteljau arithmetic visits 147 adaptive nodes and certifies the full tree whose unpruned frontier contains 16,384 cells. Taylor, fifth-root generation, and the P366 Krawczyk map remain outside the end-to-end formalization.

P395 proves by exact rational Sturm arithmetic that the degree-ten derivative of the P380 dual has exactly nine roots in $(0, 1)$, each isolated in a box of width at most $2^{-50}$. Six are near active constraints and three are inactive. Exact interval evaluation at all seven P366 primal atom boxes gives strictly positive complementary slack. The smallest lower slack is $5.54 \times 10^{-10}$. Thus the frozen P366/P380 pair is near-contact but not an exact complementary pair. P395 then solves a new simultaneous 25-equation contact system. Its residual is $3.42 \times 10^{-28}$ and its objective is 0.7073534677231137, but its Jacobian condition number is $2.73 \times 10^{15}$; interval certification remains open.

P396 isolates the strict-kernel injectivity proof as a reduction to Lindemann–Weierstrass linear independence of four exponentials. P397 then refutes exactly one proposed source class for the damping completion atom: it is not a multiplicative character of additive distance.

P398 searches for distant compensating aliases from four starts in a wide 66-loss/66-phase chart. Every fit returns near the canonical origin and no distant alias candidate is found; the best transfer defect is $4.32 \times 10^{-6}$. This is a nonexhaustive atlas, not a global theorem.

P400 optimizes permutation-symmetric pure codebooks under dephasing and improves the better product/GHZ baseline by up to 0.06138. P401 performs a full twelve-mode one-use search and finds no improvement over the extremal two-mode input in the supplied isotropic noise. P402 solves twelve finite-grid clock maximin designs; the best worst-case feasible distance is 0.99999047.

P403 provides both a custody-aware executable experiment contract and a separate 50,000-event moment-reconstruction integration test. Both pass while physical evidence remains false. P404 proves the JSR-specific law $I(S; X) = H_2(q_-)$, equal to 0.874517 bits here. P405 constructs a conditional sign-phase encoder but refutes its interpretation as an increasing negativity monotone or non-premise selector. P406 proves a unique outer scale only after the quarter-tail normalization 1/4 is supplied.

P399 and P407–P410 remain external evidence gates. The strongest surviving interpretation is a rigorous finite spectral and signed-moment framework with an executable operational interface, whose conversion into physics still requires physical references, standards, apparatus, and independent data.

# Confidence convention

| Label | Meaning |
|---|---|
| [Proven] | Mathematical proof, exact arithmetic, kernel-checked terminal predicate, or exhaustive finite result. |
| [Strong evidence] | Reproducible bounded or high-precision computation with explicit limitations. |
| [Conditional] | Requires a named encoding, apparatus law, calibration, clock, standard, or axiom. |
| [Refuted] | The declared proposition fails in its stated class. |
| [Blocked by external evidence] | Independent data, hardware, custody, standards, or unblinding is required. |

# Chapter 1

# Scope and guardrails

## 1.1 Kernel split

The legacy intermediate kernel is

$$K_{\text{legacy,ont}}(d) = 4\ln 2\frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d},$$

whereas the strict working kernel is

$$K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

They are not silently identified. P397 audits only one source class for the already target-defined attenuation completion atom. It neither derives the strict kernel from the legacy kernel nor transfers any legacy physical role.

## 1.2 Ontology and observer boundary

The internal order remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The nadsoliton is primordial information in a solitonic state, with no lower informational substrate. A common self-adjoint generator can generate both $e^{-itA}$ and $e^{-tA}$, but functional calculus does not itself specify a physical state, clock, apparatus, environment, instrument, or record.

## 1.3 Open boundaries preserved

This report does not claim:

- QW-2191 discharge or a non-premise selector;
- a source-independent complete legacy-to-strict map;
- transfer of legacy electroweak, electromagnetic, or hierarchy roles;
- internal length, time, action, mass, or energy standards;
- a calibrated twelve-state laboratory realization or vertex POVM;
- a unit-bearing nonproxy $L_{\text{total}}$;
- Standard Model, gravity, or Theory-of-Everything closure.

# Chapter 2

# Executive results

| Program | Principal result | Status |
|---|---|---|
| P394 | Complete Bernstein subdivision algorithm reflected by Lean | Proven; remaining generators open |
| P395 | Exact old-dual audit plus 25-equation KKT candidate | Proven audit; strong candidate |
| P396 | Four-exponential injectivity reduction | Proven externally; local LW formalization open |
| P397 | Additive-character damping source excluded | Proven in one class |
| P398 | Four-start wide-box alias atlas; no distant candidate | Moderate evidence |
| P399 | Independent photonic pilot absent | Blocked |
| P400 | Symmetric codebook improves product/GHZ baselines | Strong evidence |
| P401 | Full twelve-mode search finds no one-use multimode gain | Strong evidence |
| P402 | Twelve clock-error maximin designs | Proven finite-grid result |
| P403 | Custody contract plus moment validator | Proven computation; physical execution blocked |
| P404 | JSR negativity-to-information law | Proven within JSR |
| P405 | Sign-phase identity; not selector or negativity monotone | Proven; refuted interpretation |
| P406 | Unique scale after supplied quarter-tail law | Conditional; canonicity refuted |
| P407–P410 | QW, standards, reservoir, and EW records absent | Blocked |

# Chapter 3

# P394: algorithm-reflected Bernstein feasibility

P394 moves the complete depth-fourteen Bernstein subdivision algorithm into Lean. Each cell stores twelve exact integer numerators and a positive common denominator. De Casteljau splitting at one half and the denominator update are recomputed exactly. A node is pruned only when all its Bernstein coefficients lie in $[-1, 0]$.

The full unpruned frontier contains 16,384 cells. Adaptive execution visits 147 nodes and the Lean kernel proves that the recursive certificate returns true. This is algorithm reflection, not merely terminal predicate checking.

A companion source checks twenty-two exact terminal fifth-root brackets, but their generation remains Python-side. The trust boundary is:

| **Layer** | **Status** |
|---|---|
| Complete Bernstein recursive algorithm | Lean-kernel checked |
| Terminal fifth-root inequalities | Lean-kernel checked |
| Generation of fifth-root endpoints | Python implementation |
| Moment cosine Taylor/Lagrange bounds | Python implementation |
| P366 Krawczyk map and interval propagation | Python implementation |
| Lindemann–Weierstrass theorem | External theorem |

P394 fully reflects one algorithm while explicitly leaving the other generators open.

# Chapter 4

# P395: exact contact topology

Let $p(x) = \sum_{j=0}^{11} a_j x^j$ be the frozen rational P380 dual polynomial, already certified to lie in $[-1, 0]$ on $[0, 1]$. The rational Sturm chain of $p'$ proves

$$\#\{x \in (0,1) : p'(x) = 0\} = 9.$$

Every root lies in a rational box of width at most $2^{-50} = 8.88 \times 10^{-16}$.

| No. | $x$ midpoint | $p(x)$ midpoint | Boundary distance |
|---|---|---|---|
| 1 | 0.0198303330 | -0.9999999793 | $2.07 \times 10^{-8}$ |
| 2 | 0.1294250512 | $-5.08 \times 10^{-9}$ | $5.08 \times 10^{-9}$ |
| 3 | 0.2003754633 | -0.0588238886 | 0.0588 |
| 4 | 0.2950459798 | $-3.22 \times 10^{-10}$ | $3.22 \times 10^{-10}$ |
| 5 | 0.4144591497 | -0.0835768539 | 0.0836 |
| 6 | 0.5266937250 | $-1.80 \times 10^{-9}$ | $1.80 \times 10^{-9}$ |
| 7 | 0.6905262847 | -0.3540308952 | 0.3540 |
| 8 | 0.8140624992 | $-7.54 \times 10^{-9}$ | $7.54 \times 10^{-9}$ |
| 9 | 0.9418431582 | -0.9999999395 | $6.05 \times 10^{-8}$ |

Six points are near active boundaries and three are inactive. Exact rational interval evaluation on the five Krawczyk node boxes and two fixed nodes gives strictly positive complementary slack at every primal atom. The smallest certified lower slack is

$$5.53659989375616 \times 10^{-10}.$$

The frozen primal and dual therefore cannot be an exact complementary pair, although their objective gap remains below $3.05 \times 10^{-8}$.

P395 then moves all six interior nodes, seven weights, and twelve dual coefficients simultaneously. The 25-equation KKT candidate has contact pattern

$$(-1, 0, 0, 0, 0, -1, 0),$$

objective 0.7073534677231137, and maximum 80-digit residual $3.42 \times 10^{-28}$. Its smallest double-precision Jacobian singular value is $4.99 \times 10^{-13}$ and its condition number is $2.73 \times 10^{15}$. Severe ill-conditioning prevents theorem status: interval Krawczyk inclusion and global Bernstein feasibility remain required.

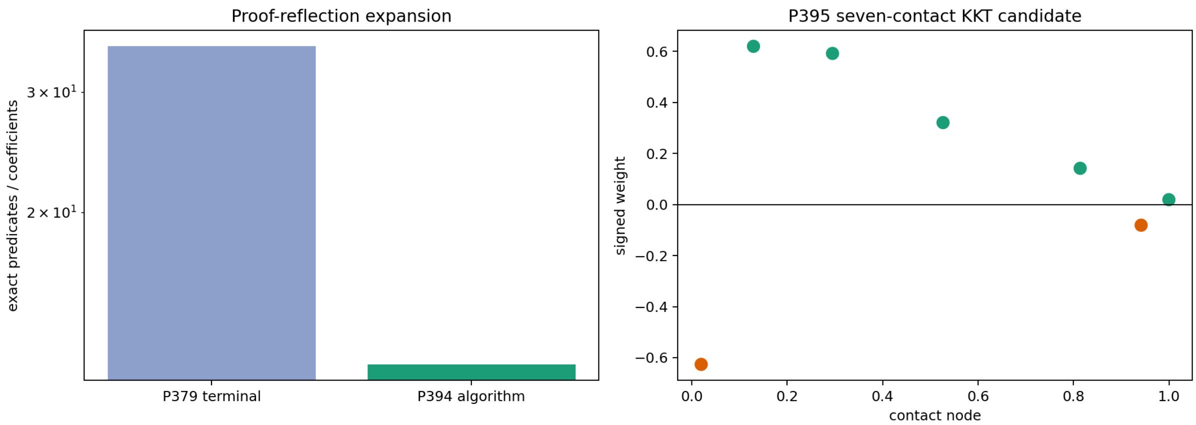


Figure 4.1: P394 algorithm reflection and the P395 seven-contact KKT candidate.

# Chapter 5

# P396: injectivity and its formal boundary

For integer $d \geq 0$, define

$$x_d = \frac{743d + 650}{4000}, \qquad a_d = \frac{1}{1 + d^{9/5}}.$$

If $K_{\mathrm{strict}}(d) = K_{\mathrm{strict}}(e)$, then

$$a_d e^{ix_d} + a_d e^{-ix_d} - a_e e^{ix_e} - a_e e^{-ix_e} = 0.$$

For $d \neq e$, the four exponents $ix_d, -ix_d, ix_e, -ix_e$ are distinct algebraic numbers, and the coefficients are nonzero algebraic numbers. Lindemann–Weierstrass linear independence forbids the relation. Hence the strict kernel is injective on $\mathbb{N}_0$.

The local Lean artifact checks the final structural implication once the independence premise is supplied. It does not locally mechanize transcendence theory. Full reflection requires a formal library for algebraicity, distinctness, and Lindemann–Weierstrass.

# Chapter 6

# P397: one damping-source class fails

The target-defined damping completion is

$$C(d) = \frac{1 + d/100}{1 + d^{9/5}}.$$

If it were a multiplicative character of additive distance, it would satisfy $C(2) = C(1)^2$. Since

$$C(1) = \frac{101}{200}, \qquad C(2) = \frac{51/50}{1 + 2^{9/5}},$$

equality would force

$$2^{9/5} = \frac{30599}{10201}.$$

Taking fifth powers yields the exact contradiction

$$30599^5 - 512\,10201^5 = -2973187286702487435951 3 \neq 0.$$

Thus the entire positive additive-character source class is excluded by one exact witness. This is not a no-go for nonlinear RG cocycles, nonlocal memory, or another new typed source.

# Chapter 7

# P398–P399: photonic quotient and hardware boundary

After quotienting exact $2\pi$ phase periodicity, P398 starts four searches in the small chart

$$\ell_j \in [0, 0.03], \qquad \varphi_j \in [-0.1, 0.1].$$

The optimizer may move in

$$\ell_j \in [0, 0.5], \qquad \varphi_j \in [-\pi, \pi].$$

All fits return to parameter norm of order $10^{-6}$. No distant alias meets the $10^{-8}$ transfer gate; the best raw transfer defect is $4.32 \times 10^{-6}$. Because the nonconvex optimizer does not reach the exact canonical zero to machine precision, the correct reading is a bounded atlas with no alias found, not global injectivity.

P399 remains blocked because no independent pilot with raw timestamps, complex transfer tomography, calibration hash, custody, and authorized unblinding is present.

# Chapter 8

# P400: noise-adapted symmetric codebooks

For $n$ uses, a permutation-symmetric pure input is parameterized by Hamming sector probabilities $p_0, \ldots, p_n$:

$$|\psi_p\rangle = \sum_{k=0}^{n} \sqrt{\frac{p_k}{\binom{n}{k}}} \sum_{|x|=k} |x\rangle.$$

Independent eigenbasis dephasing multiplies a matrix element by $q^{d_H(x,y)}$. Across 64 settings with $n \leq 4$, the optimized symmetric codebook improves the better product/GHZ baseline by as much as

$$0.0613792.$$

The maximum remaining gap to the hybrid adaptive upper bound is

$$0.4926401.$$

This is a nonconvex, ancilla-free, parallel, loss-free lower bound, not an unrestricted adaptive optimum.

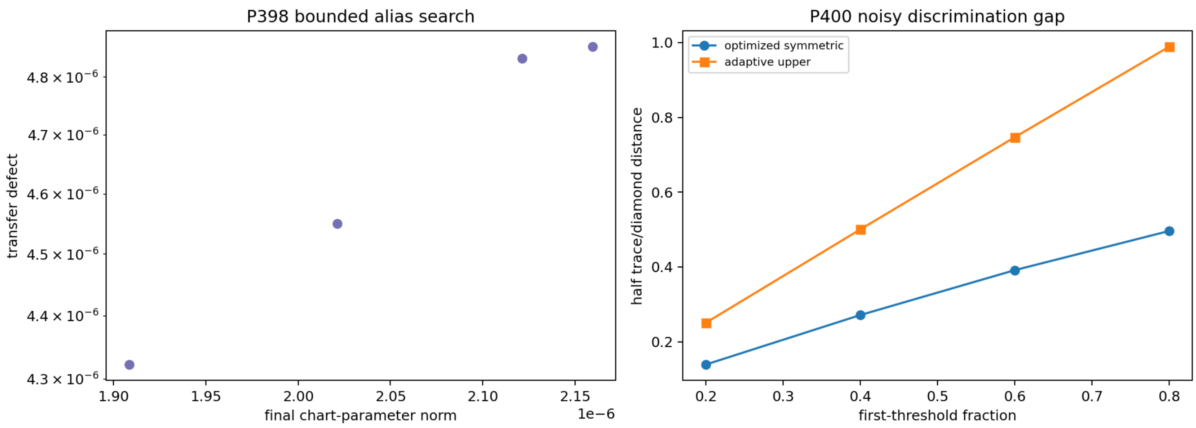


Figure 8.1: P398 bounded alias search and P400 symmetric-codebook gap.

# Chapter 9

# P401–P402: twelve modes and clock maximin design

## 9.1 Full twelve-mode one-use search

The strict and amplitude-absorbed legacy circulant generators share a Fourier basis. The relative generator is Fourier diagonal to defect $5.07 \times 10^{-15}$. P401 optimizes an arbitrary probability vector over all twelve modes for uniform off-diagonal dephasing $q$ and heralded survival $\eta$.

Across 36 settings, no input improves on the equal superposition of the two extremal relative-generator modes beyond reported numerical precision. This negative result supports, but does not prove, one-use extremal-two-mode optimality. Multiple adaptive uses and apparatus calibration remain open.

## 9.2 Clock-error maximin schedules

For every $n = 1, \ldots, 4$ and $q \in \{0.5, 0.75, 1\}$, P402 searches 1,501 nominal times on the first branch. With $\delta\tau \in [-0.001, 0.001]$, the objective is the smaller feasible discrimination at the two endpoints. Twelve designs are solved; the best worst-case value is 0.9999904711.

This is exact only for the finite grid and supplied tube/noise model. It does not create a physical time unit.

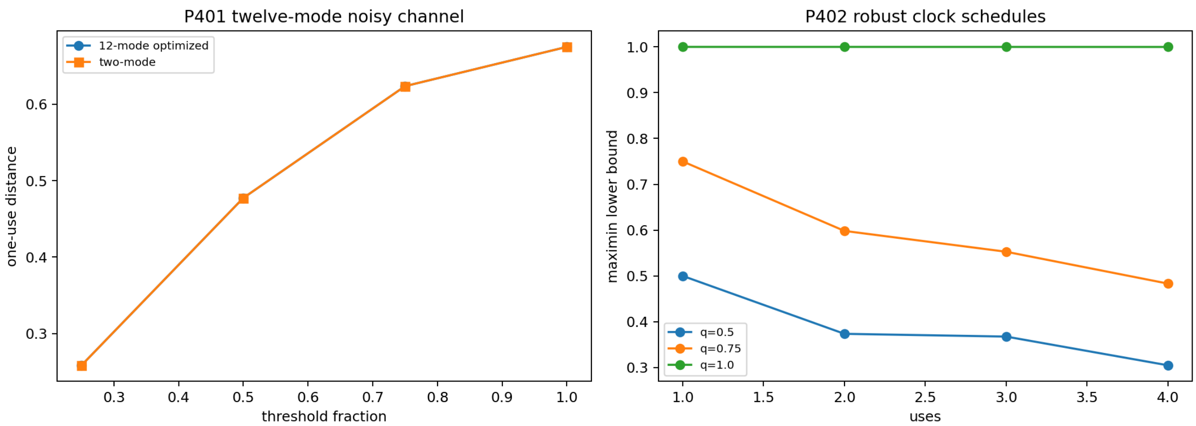


Figure 9.1: P401 twelve-mode search and P402 clock maximin schedules.

# Chapter 10

# P403: executable Jordan Sampling Realization

For a finite signed measure $\mu = \sum_i w_i \delta_{x_i}$, let

$$V = \sum_i |w_i|, \qquad q_i = \frac{|w_i|}{V}, \qquad s_i = \operatorname{sgn}(w_i).$$

The event score for moment order $k$ is $Y_k = V s_i x_i^k$, and

$$\mathbb{E}_q Y_k = \sum_i w_i x_i^k.$$

All preparation probabilities are ordinary and nonnegative. The sign is a classical importance-sampling register, not negative probability.

P403 freezes two complementary executable layers. The custody-aware transfer contract consists of:

- `FIN_P403_JSR_Executable_Spec.json`;
- `fin_p403_jsr_validator.py`.

It requires nine event fields, eleven manifest fields, three distinct roles, hash-before-unblinding, one analysis run, and no model repair after failure.

The moment-reconstruction integration test consists of:

- `FIN_P403_Jordan_Sampling_Protocol.json`;
- `FIN_P403_Jordan_Sampling_Synthetic_Events.csv`;
- `fin_p403_moment_validator.py`;
- `FIN_P403_Jordan_Sampling_Validation.json`.

Each event contains event id, run id, timestamp tick, atom, node, and sign. The validator checks schema, unique identifiers, strict timestamp order, frozen atom identity, normalized probabilities, raw-record hash, empirical atom frequencies, and twelve moment estimates.

For 50,000 synthetic events, the maximum moment $z$-score is 1.583. The validation passes. The field `physical_evidence_admitted` is false.

The synthetic bundle is an integration test, not laboratory evidence. The structural self-test checks logic, not the truth of custody metadata. A real execution must add independent preparation, apparatus, detector calibration, custody, hold-out, and unmodified failure reporting.

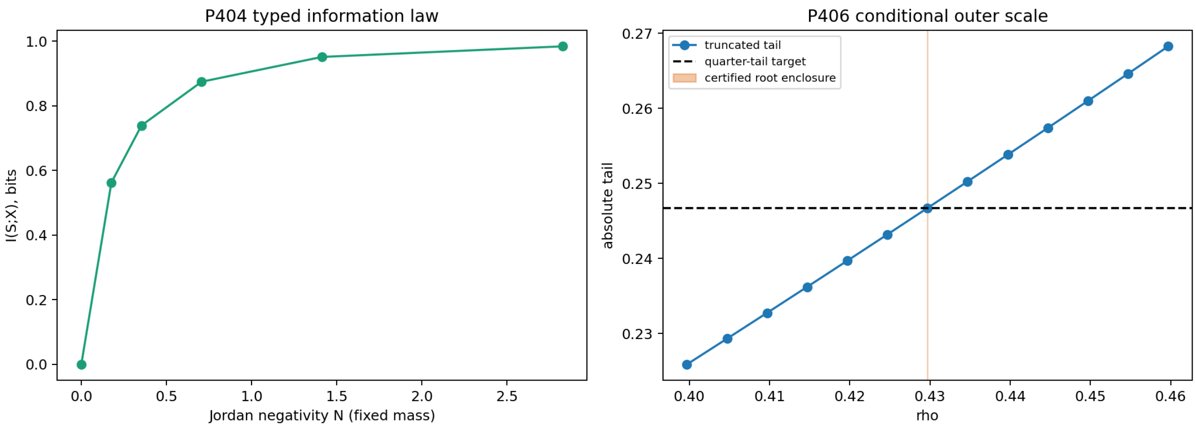


Figure 10.1: P404 JSR information law and the P406 conditional outer section.

# Chapter 11

# P404–P405: JSR information and conditional phase

## 11.1 Negativity-to-information law inside JSR

For signed mass $m > 0$ and Jordan negative mass $N \geq 0$,

$$q_- = \frac{N}{m + 2N}.$$

Because the atom determines its sign,

$$I(S; X) = H_2(q_-).$$

At fixed $m$, this increases with $N$ until $q_- = 1/2$, and data processing gives $I(S; Y) \leq I(S; X)$. For the seven-atom object:

$$m = 0.9868259032, \quad N = 0.7073534684, \quad q_- = 0.2945424925, \quad I(S; X) = 0.8745170155 \text{ bits}.$$

This is specific to the JSR encoding and does not turn Shannon information into thermodynamic entropy.

## 11.2 Conditional sign-phase encoding

Encode $s = +1$ as $|+\rangle$ and $s = -1$ as $|-\rangle$. The mixture obeys

$$C_{l_1}(\rho_{\text{phase}}) = \text{TV}(p_S, \mathsf{R}p_S) = |1 - 2q_-|.$$

The value is 0.4109150151 for the certified object. At fixed positive mass it decreases as negativity increases, so it is sign bias rather than an increasing negativity monotone. Swapping the encoded basis reverses polarity without changing magnitude. The basis/orientation is supplied and QW-2191 remains open.

# Chapter 12

# P406: quarter-tail normalized outer section

Define

$$F(\rho) = \sum_{d\geq 1} |K_{\mathrm{strict}}(d)|\rho^d.$$

P406 adds exactly one boundary law:

$$F(\rho) = \frac{|K_{\mathrm{strict}}(0)|}{4}.$$

The positive series is continuous and strictly increasing. A 160-term truncation and

$$0 \leq R_{160}(\rho) \leq \frac{\rho^{161}}{161^{9/5}(1-\rho)}$$

give the unique conditional solution

$$\rho \approx 0.4296970877901551$$

with computed enclosure width $2.30 \times 10^{-63}$. The factor $1/4$ is an added, dimensionless normalization. It is not source-derived and creates no length, action, mass, energy, or SI unit.

# Chapter 13

# External gates P407–P410

No artifact satisfies the required provider, registrar, calibration-hash, raw-record-hash, freeze-time, analyst, and unblinding fields for:

- P407: independent QW hold-out;
- P408: dimensional standards;
- P409: reservoir process tomography;
- P410: electroweak blind test.

These are deliberately unfilled empirical slots. They require independent people, apparatus, standards, and records.

# Chapter 14

# New theoretical objects

| Object | Name | Exact boundary |
|---|---|---|
| O131 | Algorithm-Reflected Bernstein Certificate | Other generators remain open |
| O132 | Seven-Contact KKT Extremal | Interval/global certification open |
| O133 | Transcendence Dependency Interface | Reduction, not local transcendence library |
| O134 | Additive-Character Damping Obstruction | One source class only |
| O135 | Bounded Photonic Alias Atlas | Four searches, not global/hardware |
| O136 | Noise-Adapted Symmetric Codebook | Restricted lower bound, not optimum |
| O137 | Twelve-Mode Noisy Discrimination Simplex | One use only; no lab |
| O138 | Noisy Clock Maximin Schedule | Frozen finite design only |
| O139 | JSR Executable Experiment Contract | Software/custody schema, not lab evidence |
| O140 | JSR Sign-Information Transformation | JSR-specific, not thermodynamics |
| O141 | Binary Sign-Phase-Asymmetry Encoder | Not negativity monotone or selector |
| O142 | Quarter-Tail Normalized Outer Section | Conditional scale, not internal unit |

# Chapter 15

# Falsification summary

The round attacked its most attractive interpretations.

1. Exact contact for the frozen P366/P380 pair is refuted by positive slack.
2. Complete local formalization is refuted by the explicit trust matrix.
3. Additive-semigroup character origin of $C_{\mathrm{damp}}$ is refuted exactly.
4. Product/GHZ completeness is refuted by a better symmetric codebook.
5. One-use twelve-mode advantage is not found under the supplied noise.
6. Synthetic JSR is prevented from becoming physical evidence by its admission flag.
7. Encoded coherence is refuted as an increasing negativity monotone.
8. Sign-phase encoding does not become a selector without a reference orientation.
9. The dilation orbit still does not select its own scale; P406 adds 1/4.
10. No program supports full bridge closure or legacy role transfer.

# Chapter 16

# Present interpretation of FIN

The strongest interpretation surviving the round is:

> FIN currently defines a finite spectral-operator and signed-moment framework with dual unitary/diffusive functional calculi, a rigorous radial graph geometry, and at least one explicit positive-probability operational realization. Its transition to physics is conditional on separately supplied reference structures—orientation, phase, clock, scale, apparatus, and empirical custody—not a consequence of the spectral operator alone.

The same self-adjoint generator explains why

$$U_t = e^{-itA}, \qquad P_t = e^{-tA}$$

coexist mathematically. It does not choose which dynamics is physically implemented, calibrate $t$, prepare a state, or specify an instrument. P403 answers these questions for one declared signed-moment sampling task at the mathematical/software level. P406 answers scale selection only after an external quarter-tail normalization is supplied.

# Chapter 17

# Recommended Programs P411–P427

| P | Study | Success criterion | Prior |
|---|---|---|---|
| 411 | Mechanize cosine/Taylor intervals | All rational cosine bounds derived in proof assistant | 0.70 |
| 412 | Solve exact P395 contact equations | Exact optimizer, uniqueness, or certified nonuniqueness | 0.55 |
| 413 | Formalize global injectivity with transcendence library | Kernel-checked full P381 theorem | 0.35 |
| 414 | Test one normalized nonlinear RG cocycle source | Source-derived damping atom or exact class no-go | 0.30 |
| 415 | Global interval photonic identifiability | Full-box nonsingularity or explicit alias | 0.45 |
| 416 | Independent photonic pilot | Admitted calibrated raw transfer record | 0.30 |
| 417 | Noisy two-mode comb SDP | Matching primal and dual certificate | 0.55 |
| 418 | Erasure-corrected entangled strategy | Strict improvement with resource accounting | 0.50 |
| 419 | Full twelve-mode noisy comb | Certified gap below $10^{-3}$ | 0.30 |
| 420 | Robust design with real clock anchor | Frozen SI-to-$\tau$ calibration | 0.25 |
| 421 | External execution of P403 | Independent raw events pass frozen validator | 0.35 |
| 422 | Finite-sample optimal JSR estimators | Proven variance reduction without bias | 0.70 |
| 423 | Reference-frame theory for sign-to-coherence | Exact monotone and reference cost | 0.60 |
| 424 | One strict oriented-record source outside scalar classes | Nonzero nonconventional odd source or bounded no-go | 0.15 |
| 425 | Compare scale boundary laws | Equivalence/non-equivalence theorem | 0.65 |
| 426 | QW/standards/reservoir admission campaign | One preregistered gate accepted unchanged | 0.20 |
| 427 | Blinded EW test after bridge closure only | Frozen prediction and independent unblinding | 0.05 |

The preferred sequence is

$$\boxed{\text{P411} \to \text{P412} \to \text{P413} \to \text{P417} \to \text{P421}.}$$

P411–P413 reduce mathematical trust and settle extremal geometry. P417 addresses the largest operational theorem gap. P421 is the shortest honest route from an executable specification to an external empirical record. P414 must stop after the named cocycle class. P424 is admissible only after an explicit new inversion-odd formula is supplied; generic selector replay is not.

# Chapter 18

# Reproducibility

The round is reproduced by:

```
python3 fin_programs_394_410.py
python3 fin_p403_jsr_validator.py --self-test \
  --spec FIN_P403_JSR_Executable_Spec.json
python3 fin_p403_moment_validator.py \
  FIN_P403_Jordan_Sampling_Protocol.json \
  FIN_P403_Jordan_Sampling_Synthetic_Events.csv
python3 -m unittest -v test_fin_programs_394_410.py
```

The fixed random seed is 20260765. Exact rational arithmetic is used for Sturm counts, root isolation, fifth-root inequalities, the damping-character counterexample, and complementary-slack intervals. Floating-point linear algebra is used for photonic inversion and matrix exponentials and is labelled as such.

# Chapter 19

# Conclusion

P394–P410 do not establish FIN as physical theory. They make its boundary sharper. The oscillatory certificate now has exact critical topology and a proved nonzero contact mismatch. The strict radial theorem has a clean formal interface to transcendence theory. A tempting semigroup origin of nonlinear damping is impossible. The photonic alias atlas finds no distant alias but does not prove global identifiability. A symmetric codebook beats both product and GHZ baselines, while the full twelve-mode one-use search finds no advantage over the extremal two-mode state. Conditional clock maximin schedules exist. Most importantly, the signed-moment resource now has a custody-aware executable contract and an event-level moment validator.

The remaining bridge to physics is not another reinterpretation of the same operator. It is the controlled addition or empirical determination of physical preparation, clock, phase/orientation reference, dimensional standard, apparatus response, and independent record. Until then, FIN's strongest defensible status is a rigorous finite spectral and operational mathematical framework with well-specified experimental interfaces.